Bangladesh University of Engineering and Technology

Department of Computer Science and Engineering

Course: CSE 206

Digital Logic Design Sessional

Experiment No. 10

Topic: **Registers**

Design and Implement the Following Problems:

1. Design and implement a 4-bit universal shift register. You can use IC 7474 (Dual D flip-flop) for this problem. Your approach must ensure this :

Table 1: Control Signal Operation Table

S0	S1	Expected Output
0	0	Hold
0	1	Parallel Load
1	0	Shift Right
1	1	Shift Left

Questions:

1. Design a 4-bit serial-in parallel-out shift register. Your register should perform two operations: serial load and rotate. You can use a 2×1 MUX for this purpose. Implement your register keeping in mind the following specification:
   - Serial load from the left.
   - Rotate left the leftmost 3 bits only (e.g., 1010 becomes 0110 after one rotation, 1100 after two rotations).
2. Write a short note on IC 7491A.

Report:

1. Problem specification.
2. Required instruments.

3. Circuit diagram with PIN number.

4. Answer to the questions.